

MRI Scanner Operations and Safety Manual — Radiology Suite B

Sunrise Medical Center

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Distribution: All Sunrise Medical Center Staff

1. Scope of This Manual

This manual governs the safe operation of the 3.0 Tesla MRI scanner installed in Radiology Suite B at Sunrise Medical Center. It covers pre-scan safety screening, patient positioning, contrast administration, emergency quench procedures, and routine maintenance. All MRI technologists must complete annual competency verification against this manual.

2. Zone Classification

The MRI suite is divided into four zones of increasing magnetic field risk, consistent with ACR guidance. Zone I is the public area, Zone II is the transition/screening area, Zone III is restricted to screened personnel and patients, and Zone IV is the scanner room itself, accessible only when the screening checklist has been fully completed and verified by a second technologist.

3. Pre-Scan Safety Screening

Every patient and any accompanying individual entering Zone III or IV must complete the MRI Safety Screening Form (SMC-RAD-F12) in writing, verified verbally by the technologist. No exceptions are permitted for screening, including for staff, emergency responders, or family members.

- Cardiac pacemakers, implanted defibrillators, and most older neurostimulators are absolute contraindications unless documented as MR Conditional with specific parameters.
- Cochlear implants, certain aneurysm clips, and metallic foreign bodies near the eye require case-by-case radiologist review before scanning.
- All ferromagnetic objects (oxygen tanks, non-MR-safe wheelchairs, keys, badges, hairpins) must be removed before Zone III entry.
- Pregnant patients require radiologist consultation prior to scanning, with contrast administration avoided unless clinically essential.

4. Contrast Administration

Gadolinium-based contrast agents are administered only after renal function (eGFR) has been verified within the preceding 30 days per policy SMC-RAD-114. Patients with eGFR below 30 mL/min/1.73m² require

radiologist approval and informed consent discussing the risk of nephrogenic systemic fibrosis before proceeding.

An emergency contrast reaction kit and code cart access must be confirmed present and unexpired before any contrast-enhanced study begins.

5. Emergency Quench Procedure

In the event of a ferromagnetic projectile emergency or fire requiring immediate field shutdown, only the emergency quench button (located outside Zone IV, clearly marked in red) may be used to rapidly evacuate the liquid helium and eliminate the magnetic field. This is an irreversible action costing significant repair time and expense, and must only be used for genuine life-threatening emergencies — never for routine equipment issues.

- 1 Evacuate all personnel and the patient from Zone IV if physically able to move safely.
- 2 Activate the emergency quench button only if the patient or a person is pinned by a ferromagnetic object and cannot otherwise be freed.
- 3 Notify Facilities Engineering and the on-call Radiology Manager immediately.
- 4 Do not re-enter the scanner room until Facilities confirms oxygen levels are safe, as helium venting displaces breathable air.

6. Routine Maintenance Schedule

Task	Frequency	Performed By
Helium level check	Weekly	Field Service Engineer
Gradient coil calibration	Quarterly	Field Service Engineer
Quench pipe inspection	Annually	Facilities Engineering
RF coil function test	Daily (start of shift)	MRI Technologist
Emergency stop button test	Monthly	Chief Technologist

7. Staff Training Requirements

All technologists assigned to the MRI suite must complete initial MR Safety Officer-led orientation, pass a written competency exam, and complete supervised scanning of at least 20 studies before working independently. Annual refresher training is mandatory and tracked by Human Resources.

8. Related Documents

SMC-RAD-F12 MRI Safety Screening Form; SMC-RAD-114 Gadolinium Contrast Administration Policy; SMC-FAC-033 Zone IV Access Control Procedure; SMC-RAD-102 CT Scanner Operations Manual.

Appendix A: Frequently Asked Questions

Q: Can a patient with an old, unknown implant type be scanned? A: No. Scanning must be deferred until the specific implant model is identified and confirmed MR Safe or MR Conditional; operative reports or

manufacturer cards should be requested before proceeding.

Q: Are hospital staff required to be screened every time they enter Zone IV? A: Yes, every entry requires screening confirmation, even for staff who have entered previously that same day, since carried items can change between visits.

Q: What should be done if a patient becomes anxious or claustrophobic during the scan? A: The technologist can pause the scan, communicate via the intercom, and offer the emergency squeeze-ball alert; sedation may be arranged in advance for known claustrophobic patients per physician order.

Q: How often is the MRI safety screening form updated? A: The form is reviewed annually by the MR Safety Officer and updated immediately if new implant guidance is issued by the ACR or FDA.

Appendix B: Common Scenarios

Scenario — Unconscious trauma patient needing MRI: When history cannot be obtained directly, staff must exhaust all available sources including family, EMS report, prior imaging, and visible surgical scars before making a screening determination, and involve the radiologist in the final risk decision.

Scenario — Oxygen tank brought into Zone III by transport staff: Only non-ferromagnetic, MR-safe oxygen tanks are permitted past the Zone II threshold; standard steel tanks must be exchanged at the Zone II boundary before the patient proceeds further.

Revision History

Version	Date	Description
1.0	May 5, 2020	Initial release for 3.0T scanner installation
1.3	Oct 22, 2025	Updated gadolinium eGFR threshold and screening form revision